Setif1-Ferhat Abbas University *Matter Sciences*
Faculty of sciences

Maths2

5^{th} guided works

Exercise 1

Determine which of the sets E_1, E_2, E_3 are vector subspaces of $\mathbb{R}^3$
$E_1 = \{(x, y, z) \in \mathbb{R}^3 : 3x - 5y = 0\}$
$E_2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$
$E_3 = \{(x, y, z) \in \mathbb{R}^3 : x + y = 0 \text{ and } z = 0\}$

Exercise 2

Show that the set $S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$ spans (generates) $\mathbb{R}^3$

Exercise 3

1. Determine whether the following vectors in $\mathbb{R}^3$ are linear Independent or linear Dependent :

$$V_1 = (1, 2, 3), V_2 = (0, 1, 2), V_3 = (-2, 0, 1)$$

2. Determine whether the following second-degree polynomials are linear Independent or linear Dependent :

$$P_1 = 1 + x - 2x^2, P_2 = 2 + 5x - x^2, P_3 = x + x^2$$

Exercise 4

We consider two vetcor subspaces E, G of $\mathbb{R}^3$ defined by :
$E = \{(x, y, z) \in \mathbb{R}^3, x = 0, y = z\}$
$G = \{(x, y, z) \in \mathbb{R}^3, y = 0\}$.
Give their dimensions.

Exercise 5

We consider an application $f : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by :

$$f(x, y, z) = (x + y, x - z, y - z)$$

1) Show that f is linear.
2) Determine $Kerf$ (kernel of f) and Imf (image of f).